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Predisposing Factor Identification and Multi-Class Grading of Diabetes Mellitus Using Machine Learning

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Abstract

Diabetes mellitus is a persistent global health burden, and understanding its predisposing factors is essential for prevention and early risk assessment. This study presents a machine learning framework for diabetes prediction and multi-class grading using the CDC's BRFSS 2015 "diabetes_012" dataset, consisting of 253,680 adult respondents and 21 behavioral, clinical, and demographic variables. The target outcome includes three categories: no diabetes or gestational only, prediabetes, and diabetes characterized by substantial class imbalance. Data preprocessing involved removal of duplicate entries, detection of outliers in key continuous variables using z-score thresholds ($|z| > 3$), and z-score normalization of numeric predictors. The dataset was partitioned into stratified training (70%) and test (30%) sets, and the SMOTE algorithm was applied to the training set to correct class imbalance. To identify the most influential predisposing factors, five complementary feature selection approaches were used: Logistic Regression coefficients, Recursive Feature Elimination, Random Forest feature importance, Mutual Information, and Chi-square statistics. The intersection of top-ranked features across these methods produced a compact set of consistent predictors: high blood pressure (HighBP), high cholesterol (HighChol), body mass index (BMI), self-rated general health (GenHlth), and age. These variables represent core cardiometabolic and self-reported health dimensions with strong relevance to diabetes mellitus. Using only this compact factor set, several supervised machine learning models were developed and compared, including k-Nearest Neighbors, Support Vector Machine with RBF kernel, Random Forest, multinomial Logistic Regression, and XGBoost. Final model evaluation was conducted on a representative subset of 40,000 training and 10,000 test observations following SMOTE balancing. Performance assessment based on accuracy, macro-averaged F1-score, and confusion matrices demonstrated that XGBoost achieved the strongest results, with an accuracy of 83.11% and a macro F1-score of 0.42, effectively distinguishing diabetic from non-diabetic respondents and offering moderate separation of prediabetes. Overall, the findings indicate that a small, interpretable set of predisposing factors combined with XGBoost-based machine learning provides a practical, scalable framework for multi-class grading of diabetes mellitus in population-level survey data.